

CHERI FAVEL

Chemical / Process Engineer (MEng) | R&D • Hydrometallurgy • Water & Environment

Abu Dhabi, UAE | South African | UAE Resident Visa | Availability: Immediate

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PROFILE

MEng Chemical Engineer with combined academic and applied experience in process development, hydrometallurgy, electrochemical systems, mineral processing, water/tailings treatment and laboratory R&D. Experienced in experimental design, process optimisation, mass and energy balances, PFD/P&ID development, equipment-sizing support, technical reporting and analytical data interpretation. Combines industry-focused process development with two years of postgraduate R&D, published research and hands-on experience across laboratory, pilot-scale and operational environments.

CORE TECHNICAL STRENGTHS

- **Process Engineering:** Process design support; process optimisation; mass & energy balances; PFDs; P&ID interpretation/development; equipment sizing; design bases; technical reports; supplier quotation review.
- **Hydrometallurgy:** Mineral leaching; electrowinning; electrochemistry; selective precipitation; metal recovery; tailings treatment; beneficiation; recovery feasibility.
- **Water / Environment:** Water & wastewater fundamentals; UF/NF/RO exposure; carbon filtration; solid-liquid separation; water chemistry; fouling/scaling awareness; environmental engineering.
- **Laboratory / Analytical:** ICP-OES, XRF, XRD, ATR-FTIR, BET, TGA-DTG, SEM-EDS/FEG-SEM, PSA, pH analysis, batch leaching, electrochemical test systems.
- **Software / Data:** Aspen Plus/ HYSYS (foundational); Microsoft Visio; Python; GitHub; advanced Excel including VBA/macros, Power Query, PivotTables and data analysis.

ENGINEERING EXPERIENCE

Process & R&D Engineer

Mar 2021 – Jul 2022

Free Radical Process Design (FRPD) | South Africa

- Conducted technical investigations and experimental programmes supporting process development and optimisation across hydrometallurgical, electrochemical and water-treatment projects.
- Designed process units; performed mass/energy balances and process calculations; supported equipment sizing; developed PFDs/P&IDs, process descriptions, specifications, design bases and technical reports.
- Managed client-focused laboratory investigations and laboratory activities from experimental planning and resource coordination through execution, data interpretation, reporting and engineering recommendations.
- Reviewed supplier quotations from a process perspective and supported safety-aware design/procedural considerations.

SELECTED ENGINEERING & R&D PROJECTS

- **Phlogopite Leaching & Potassium Recovery (Master's Research):** Designed and executed nitric-acid batch leaching experiments for potassium recovery/fertiliser applications; developed kinetic models, determined activation parameters and characterised feed/residues using XRF, XRD, ATR-FTIR, BET, TGA and SEM-EDS. Research published in *Clays and Clay Minerals*.
- **Continuous Cobalt Electrowinning:** Supported development and evaluation of a novel continuous electrochemical recovery process.
- **Tailings Water Treatment, Peru:** Investigated Fe, Zn and Mn removal/recovery using oxidation, precipitation and electrolysis; assessed treatment performance and process feasibility.
- **Manganese Processing & Beneficiation:** Supported manganese sulphate monohydrate process development and investigated selective precipitation and electrochemical beneficiation routes.
- **Gold Recovery Reactor:** Contributed to laboratory-scale electrowinning/sludging reactor development for residual gold recovery from refinery eluate; investigated current density, voltage, flow rate and reaction time using ICP-OES, XRF and pH analysis.
- **Rotating-Cathode Zinc Removal Reactor:** Worked on an electrochemical system for zinc removal from tailings water, including scale-up considerations toward mobile-plant application.
- **Hydrotalcite-Like LDH Synthesis:** Synthesised LDH-type material from phlogopite-derived leach liquor via coprecipitation and supported characterisation using XRD, FTIR and SEM.
- **Fe/Al Precipitation from Leach Liquor:** Investigated MgO, CaCO₃ and limestone/eggshell as precipitation agents for Fe³⁺ and Al³⁺ removal.
- **Additional Engineering Projects:** Ni-MH battery metal-recovery feasibility; neodymium-plant CAPEX/OPEX analysis; galvanising process review; seawater and silver electrolysis; packed-bed pre-reactor design for reactive distillation; renewable-energy residential HVAC concept.

Vessel Operations & Engineering Support

May 2023 – Apr 2025

Research Vessel Rosa | United States & Bahamas

- Supported vessel maintenance, technical documentation and practical troubleshooting across engines, pumps, plumbing, RO/water-making, IT and other marine systems while working within strict operational and safety requirements.

Postgraduate R&D Engineer / Master's Research

Jan 2020 – Dec 2021

University of Pretoria | South Africa

- Independently planned and managed a two-year laboratory research programme, from experimental design and methodology development through execution, data analysis and technical reporting.
- Designed and conducted controlled experimental programmes, evaluated research hypotheses and interpreted complex chemical and process data to identify trends and draw evidence-based conclusions.
- Applied advanced data analysis and modelling using Excel and Python, maintaining rigorous experimental records, data quality, reproducibility and traceability.
- Supervised and mentored undergraduate vacation-work research assistants, communicating project objectives, delegating experimental tasks, reviewing completed work and reports, and providing technical guidance and feedback.
- Communicated research findings through technical reports, academic presentations and scientific writing, culminating in peer-reviewed publication in *Clays and Clay Minerals*.

Teaching Assistant (Part-Time)

Feb 2019 – Jun 2021

University of Pretoria | South Africa

- Supported laboratory sessions and tutoring in particle technology, process kinetics and engineering practice/costing; supervised laboratory work, assessed students and provided technical feedback.

Laboratory Intern (Vacation Work)

Dec 2017 – Jan 2019

University of Pretoria | South Africa

- Completed 12 weeks of laboratory work on mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION

MEng Chemical Engineering

University of Pretoria, South Africa | 2021

- Published research in *Clays and Clay Minerals*; postgraduate research bursary; dissertation focused on nitric-acid leaching kinetics of phlogopite and solid-residue characterisation.

BEngHons Chemical Engineering (Distinction)

University of Pretoria, South Africa | 2020

- Recipient of Academic Honorary Colours.

BEng Chemical Engineering

University of Pretoria, South Africa | 2019

- Environmental Engineering specialisation including water/effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

CERTIFICATIONS & ADDITIONAL QUALIFICATIONS

- Operations Management Professional Certificate – Jack Welch Management Institute / LinkedIn Learning | In Progress (2026)
- LEED Green Associate Cert Prep – LinkedIn Learning | Sep 2026
- Project Management – Energy and Water Projects – PMI / LinkedIn Learning | Feb 2026
- Occupational Safety and Health: Hazard Communication – LinkedIn Learning | Feb 2026
- Python Mega Course: Developing Apps and AI Agents – Udemy | In Progress
- AI for Project Management: Managing Risk with Generative AI – NASBA / LinkedIn Learning | Feb 2026
- ChatGPT Prompt Engineering for Developers – DeepLearning.AI, Feb 2026
- Approved Engine Course (AEC 1) – Maritime Professional Training | 2024
- STCW Basic Safety Training: Fire Prevention/Firefighting, First Aid, Personal Safety & Social Responsibility | 2023

LANGUAGES

- English (Native) | Afrikaans (Proficient) | Hebrew (Basic)